

Qualification Pack



Textile Dyeing Machine Operator

Electives: Jigger/ Soft Flow Dyeing/ Package Dyeing

QP Code: TSC/Q5207

Version: 3.0

NSQF Level: 2

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TSC/Q5207: Textile Dyeing Machine Operator

Brief Job Description

A Dyeing Machine Operator is responsible for pretreatment and dyeing operations. The job requires the individual to have thorough knowledge of process flow and material flow in a processing mill for fabric production and he should know the important functions and operations of textile dyeing machines.

Personal Attributes

A Textile Dyeing Machine Operator should have good eyesight, motor skills and vision. He/she should also have good interpersonal skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TSC/N9015: Follow machine, safety, and organizational guidelines in textile sector](#)
2. [TSC/N9016: Follow teamwork, adaptability, and communication guidelines in textile sector](#)
3. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Jigger

A Jigger Machine Operator is responsible for scouring, bleaching and dyeing or any other process carried out on fabric on this machine. This job requires the individual to have thorough knowledge of process flow and material flow in a processing mill for fabric production and he should know the important functions and operations of a jigger machine.

1. [TSC/N5220: Carryout shift change, production and quality analysis at jigger machine department](#)

Elective 2: Soft Flow Dyeing

A Soft Flow Dyeing Machine Operator is responsible for scouring, bleaching and dyeing of fabric in soft flow machine. This job requires the individual to have thorough knowledge of process flow and material flow in a processing mill for fabric production and should know about the important functions and operations of a soft flow machine.

1. [TSC/N5221: Carryout shift change, production and quality analysis at soft flow dyeing department](#)

Elective 3: Package Dyeing

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A Package Dyeing Machine Operator is responsible for scouring, bleaching and dyeing of fibre/ yarn. This job requires the individual to have thorough knowledge of process flow and material flow in a processing mill for yarn production and should know about the important functions and operations of a Package Dyeing machine.

1. [TSC/N5222: Carryout shift change, production and quality analysis at package dyeing department](#)

Qualification Pack (QP) Parameters

Sector	Textile
Sub-Sector	Processing - Textiles
Occupation	Dyeing & Printing
Country	India
NSQF Level	2
Credits	25
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8154.1
Minimum Educational Qualification & Experience	Ability to read and write
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-02-TX-04096-2025-V2-TSC
NQR Version	3.0

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TSC/N9015: Follow machine, safety, and organizational guidelines in textile sector

Description

This unit provides performance criteria, knowledge and skills required to follow machine, safety and organizational guidelines in textile sector.

Scope

The scope covers the following :

- Maintaining the work area, tools and machines
- Greening and energy conservation in textile sector
- Health, safety and response to emergencies at textile sector
- Organizational standards and policies

Elements and Performance Criteria

Maintaining the work area, tools and machines

To be competent, the user/individual on the job must be able to:

- PC1.** handle materials, machinery, equipment and tools as per standard procedure
- PC2.** use appropriate material handling equipment and tools as per standard procedure
- PC3.** keep the equipment, machine and work area clean using appropriate cleaning tools as per standard procedure
- PC4.** undertake minor routine maintenance of equipment and tools as per standard maintenance procedure
- PC5.** maintain record for defective and unsafe equipment and tools
- PC6.** verify that machine guards are in place as per standard specifications
- PC7.** follow specified ergonomics for the assigned job role in textile sector
- PC8.** collect and store worn-out spare parts at specified location
- PC9.** report the condition of worn out parts as per standard procedure

Greening and energy conservation in textile sector

To be competent, the user/individual on the job must be able to:

- PC10.** segregate wastes such as recyclable, non-recyclable, hazardous as per standard protocol
- PC11.** optimize usage of material and resources including water, electricity in various tasks
- PC12.** switch off the machines and lights when not in use

Health, safety and response to emergencies at textile sector

To be competent, the user/individual on the job must be able to:

- PC13.** use Personal Protective Equipment (PPEs) like body protector, ear plugs, nose mask, head cap, etc. as per guidelines
- PC14.** identify abnormal sounds emanating from faulty or worn out machine parts and take appropriate action
- PC15.** avoid dependency on any type of intoxicants

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- PC16.** maintain social distance as per the instruction at workplace
- PC17.** report hazardous material to superiors at workplace
- PC18.** use the various appropriate fire extinguishers on different types of fires correctly
- PC19.** follow the specified steps in case of electricity failure
- PC20.** lift heavy objects using correct lifting procedures
- PC21.** recall emergency exits, safe spots, etc. of workplace
- PC22.** practice mock drills and evacuation procedures organized by industry
- PC23.** assist others to reach to safe spots in emergency situations
- PC24.** provide basic first aid for injury to peers and report to superiors
- PC25.** interpret different signs, alarms and take action appropriately
- PC26.** follow the guidelines while working in hazards atmosphere
- PC27.** assist in designing the safety plans with peers and superiors
- PC28.** follow the approved safety plans at workplace

Organizational standards and policies

To be competent, the user/individual on the job must be able to:

- PC29.** perform assigned duties as per organization's protocol within scheduled time period
- PC30.** follow organization policies, quality standards, rules and regulations for working in textile sector
- PC31.** motivate colleagues to follow operational guidelines of organization
- PC32.** wear specified uniform and follow etiquette as per standard guidelines for the textile sector
- PC33.** maintain hygienic working atmosphere as per protocol of the textile sector
- PC34.** submit lost and found articles as per standard protocol

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational standard procedures, quality standards, rules, codes, policies and safety standards in the textile sector
- KU2.** different type of tools and equipment used in textile sub- sector and their specifications and operating procedures
- KU3.** safe handling procedure of tools and equipment
- KU4.** the importance of displays and written instructions for the allocated machines
- KU5.** dos and donts specific to the assigned work responsibilities
- KU6.** protocol for minimizing the wastage of material, effort and time
- KU7.** organization's formats and procedures for reporting production, defects, faults, material/tool requisition and quality parameters and task completed for assigned job
- KU8.** schedule for cleaning and waste collection for the assigned job role
- KU9.** importance and standard procedure for disposal of soft, hard, non-hazardous and hazardous wastes and materials
- KU10.** available types of material handling equipment and handling methods used in the textile sector

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- KU11.** hazards of unsafe workplace conditions and procedures in the textile industry and methods to avoid hazards
- KU12.** various types of fire extinguishers
- KU13.** importance of stable mental condition in case of emergency
- KU14.** correct work posture and importance of ergonomics for the assigned job role
- KU15.** organizational quality systems like quality circle, 5S, ISO, SA, etc. followed in the textile sector
- KU16.** importance of following work wear standards, behavioral protocols and etiquette in the textile sector
- KU17.** importance of energy conservation through proper maintenance schedule in the textile sector
- KU18.** procedures and formats for reporting lost and found material
- KU19.** different types of alarms and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and comprehend policies in notice boards displayed in the workplace
- GS2.** minimize the resource consumption for the assigned task
- GS3.** solve basic arithmetic calculations related to assigned job role
- GS4.** recognize and differentiate colors of materials used in textile sector
- GS5.** lift specified materials for the allotted task using prescribed ergonomic position
- GS6.** exhibit motor skill required for the allotted task

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintaining the work area, tools and machines</i>	10	10	-	6
PC1. handle materials, machinery, equipment and tools as per standard procedure	-	-	-	-
PC2. use appropriate material handling equipment and tools as per standard procedure	-	-	-	-
PC3. keep the equipment, machine and work area clean using appropriate cleaning tools as per standard procedure	-	-	-	-
PC4. undertake minor routine maintenance of equipment and tools as per standard maintenance procedure	-	-	-	-
PC5. maintain record for defective and unsafe equipment and tools	-	-	-	-
PC6. verify that machine guards are in place as per standard specifications	-	-	-	-
PC7. follow specified ergonomics for the assigned job role in textile sector	-	-	-	-
PC8. collect and store worn-out spare parts at specified location	-	-	-	-
PC9. report the condition of worn out parts as per standard procedure	-	-	-	-
<i>Greening and energy conservation in textile sector</i>	7	10	-	6
PC10. segregate wastes such as recyclable, non-recyclable, hazardous as per standard protocol	-	-	-	-
PC11. optimize usage of material and resources including water, electricity in various tasks	-	-	-	-
PC12. switch off the machines and lights when not in use	-	-	-	-
<i>Health, safety and response to emergencies at textile sector</i>	32	40	-	28

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. use Personal Protective Equipment (PPEs) like body protector, ear plugs, nose mask, head cap, etc. as per guidelines	-	-	-	-
PC14. identify abnormal sounds emanating from faulty or worn out machine parts and take appropriate action	-	-	-	-
PC15. avoid dependency on any type of intoxicants	-	-	-	-
PC16. maintain social distance as per the instruction at workplace	-	-	-	-
PC17. report hazardous material to superiors at workplace	-	-	-	-
PC18. use the various appropriate fire extinguishers on different types of fires correctly	-	-	-	-
PC19. follow the specified steps in case of electricity failure	-	-	-	-
PC20. lift heavy objects using correct lifting procedures	-	-	-	-
PC21. recall emergency exits, safe spots, etc. of workplace	-	-	-	-
PC22. practice mock drills and evacuation procedures organized by industry	-	-	-	-
PC23. assist others to reach to safe spots in emergency situations	-	-	-	-
PC24. provide basic first aid for injury to peers and report to superiors	-	-	-	-
PC25. interpret different signs, alarms and take action appropriately	-	-	-	-
PC26. follow the guidelines while working in hazards atmosphere	-	-	-	-
PC27. assist in designing the safety plans with peers and superiors	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. follow the approved safety plans at workplace	-	-	-	-
<i>Organizational standards and policies</i>	16	20	-	10
PC29. perform assigned duties as per organization's protocol within scheduled time period	-	-	-	-
PC30. follow organization policies, quality standards, rules and regulations for working in textile sector	-	-	-	-
PC31. motivate colleagues to follow operational guidelines of organization	-	-	-	-
PC32. wear specified uniform and follow etiquette as per standard guidelines for the textile sector	-	-	-	-
PC33. maintain hygienic working atmosphere as per protocol of the textile sector	-	-	-	-
PC34. submit lost and found articles as per standard protocol	-	-	-	-
NOS Total	65	80	-	50

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National Occupational Standards (NOS) Parameters

NOS Code	TSC/N9015
NOS Name	Follow machine, safety, and organizational guidelines in textile sector
Sector	Textile
Sub-Sector	Generic - Textiles Handloom
Occupation	Generic - Textiles & Handloom
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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TSC/N9016: Follow teamwork, adaptability, and communication guidelines in textile sector

Description

This unit provides performance criteria, knowledge and skills required to follow team work, communication and adaptability in textile sector

Scope

The scope covers the following :

- Teamwork and communication
- Adaptability

Elements and Performance Criteria

Teamwork, trust and communication

To be competent, the user/individual on the job must be able to:

- PC1.** contribute to create a positive work environment in the team
- PC2.** carry out tasks as per instructions received from superiors
- PC3.** contribute to team work as per allocated responsibility to complete the task by using appropriate tools and methods
- PC4.** build trust with team mates and superiors
- PC5.** implement the ideas after superior's approval at work place
- PC6.** communicate clearly with the team members as per standard protocol
- PC7.** use suggested hand signs, vocal sound signals to convey the information in the production area
- PC8.** listen effectively to the ideas and concerns of the peers
- PC9.** use correct and respectful terms while communicating as per industry policy
- PC10.** express views proactively and effectively
- PC11.** make efforts to resolve difference of opinion with superiors and team members
- PC12.** report to superior for problems identified in assigned duty
- PC13.** report the daily performance to superior in prescribed manner and formats

Adaptability

To be competent, the user/individual on the job must be able to:

- PC14.** adapt to flexible work environment for the assigned task
- PC15.** adapt to work with various members of different ethnicity, gender and PwD without biases
- PC16.** consider opinions of colleagues, fitters, superiors for the assigned task
- PC17.** plan the work-routine within the limits of the responsibility
- PC18.** adopt new ideas after due approval from superior for improving the productivity

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** importance of teamwork and discipline
- KU2.** limits and responsibilities for the assigned duties in the textile sector
- KU3.** possible conflicts in the assigned job role and methods to resolve the same
- KU4.** importance of teamwork, group discussions and healthy work environment
- KU5.** importance of reporting as per the standard protocol
- KU6.** hierarchy of communication and communication etiquettes in the textile sector
- KU7.** protocol for communication with different ethnicity, gender and PwD
- KU8.** report and grievance submission formats
- KU9.** importance of hand, vocal sound signals in the textile sector

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and comprehend written instructions
- GS2.** listen effectivity for the instructions
- GS3.** coordinate with team mates for the allotted tasks
- GS4.** use and comprehend prescribed voice and hand signals in the textile production area
- GS5.** fill forms and prepare reports such as production report, material requisition forms, leave application, etc. as per standard formats

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Teamwork, trust and communication</i>	10	15	-	5
PC1. contribute to create a positive work environment in the team	-	-	-	-
PC2. carry out tasks as per instructions received from superiors	-	-	-	-
PC3. contribute to team work as per allocated responsibility to complete the task by using appropriate tools and methods	-	-	-	-
PC4. build trust with team mates and superiors	-	-	-	-
PC5. implement the ideas after superior's approval at work place	-	-	-	-
PC6. communicate clearly with the team members as per standard protocol	-	-	-	-
PC7. use suggested hand signs, vocal sound signals to convey the information in the production area	-	-	-	-
PC8. listen effectively to the ideas and concerns of the peers	-	-	-	-
PC9. use correct and respectful terms while communicating as per industry policy	-	-	-	-
PC10. express views proactively and effectively	-	-	-	-
PC11. make efforts to resolve difference of opinion with superiors and team members	-	-	-	-
PC12. report to superior for problems identified in assigned duty	-	-	-	-
PC13. report the daily performance to superior in prescribed manner and formats	-	-	-	-
<i>Adaptability</i>	5	5	-	5
PC14. adapt to flexible work environment for the assigned task	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. adapt to work with various members of different ethnicity, gender and PwD without biases	-	-	-	-
PC16. consider opinions of colleagues, fitters, superiors for the assigned task	-	-	-	-
PC17. plan the work-routine within the limits of the responsibility	-	-	-	-
PC18. adopt new ideas after due approval from superior for improving the productivity	-	-	-	-
NOS Total	15	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TSC/N9016
NOS Name	Follow teamwork, adaptability, and communication guidelines in textile sector
Sector	Textile
Sub-Sector	Generic - Textiles Handloom
Occupation	Generic - Textiles & Handloom
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

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TSC/N5220: Carryout shift change, production and quality analysis at jigger machine department

Description

This unit is about carryout shift change, production and quality analysis at Jigger machine department

Scope

The scope covers the following :

- taking charge of shift from Jigger machine operator
- handing over shift to Jigger machine operator
- carry out preparatory activities for operations in the machine
- operate the machine for the specified task as per work order
- clean the machine & carryout preventive maintenance activities
- cutting the sample after specified time /no. of turns
- developing ,soaping, washing & drying the sample
- comparing the sample with the standard

Elements and Performance Criteria

Taking charge of shift from Jigger machine operator

To be competent, the user/individual on the job must be able to:

- PC1.** come at least 10 - 15 minutes earlier to the work place
- PC2.** bring the necessary operational tools to the department
- PC3.** meet the previous shift operator and discuss with him/ her regarding the issues faced by him with respect to the quality, production, spare, safety or any other specific instruction etc
- PC4.** understand the fabric being processed & process running on the machine
- PC5.** ensure the technical details are mentioned on the job card display on the machine
- PC6.** check for the availability of the spare trolley/batching trolley for unloading the fabric
- PC7.** check the next batch to be processed is ready near the machine
- PC8.** ensure the required dyes & chemicals are already weighed & prepared
- PC9.** check the cleanliness of the machines & other work areas
- PC10.** question the previous shift operator for any deviation in the above and bring the same to the knowledge of his/ her shift superior as well as that of the previous shift
- PC11.** take over the shift from the outgoing operator in a proper manner

Handing Over Shift to Jigger machine operator

To be competent, the user/individual on the job must be able to:

- PC12.** ensure proper communication regarding fabric quality & the process running on the machine while providing the details
- PC13.** provide all relevant information regarding the stoppages or breakdown in the machine, any damage to the fabric or machine
- PC14.** ensure the empty trolley is near the machine for unloading the fabric

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- PC15.** ensure the next lot to be processed, which is already stitched & arranged properly is ready near the machine
- PC16.** ensure the required dyes & chemicals for the next lot or next process are weighed & prepared
- PC17.** get clearance from the incoming counterpart before leaving the work spot
- PC18.** report to his/ her shift superiors as well as that of the incoming shift operator in case his/ her counterpart doesn't report for the incoming shift
- PC19.** ensure ,the shift has to be properly handed over to the incoming shift operator
- PC20.** report to his/ her shift superior about the quality / production / safety issues/ any other issue faced in his/ her shift and leave the department only after getting concurrence for the same from his/ her superiors
- PC21.** collect the wastes from waste collection bags, weigh them and transport to storage area
- PC22.** ensure the machine and its work place is clean

Carry out preparatory activities for operations in the machine

To be competent, the user/individual on the job must be able to:

- PC23.** understand the task mentioned in the work order
- PC24.** ensure that the machine is empty & clean
- PC25.** load the fabric from the batch trolley in the center of the machine
- PC26.** ensure fabric is crease-less and evenly fed onto the machine
- PC27.** prepare the chemical/dye-stuff required for the batch
- PC28.** inject the chemical in the machine tank/dispenser

Operate the machine for specified tasks as per Work Order

To be competent, the user/individual on the job must be able to:

- PC29.** know the operations of the machine & the plc/ control mechanism programs
- PC30.** read & understand the process being followed to do the task
- PC31.** check the ph, temp & other parameters during the process
- PC32.** maintain even temp across the width of machine
- PC33.** maintain even & minimum tension on the fabric
- PC34.** drain the machine once sample is approved
- PC35.** unload the fabric from the machine after completion of the process

Clean the machine on a regular basis and carryout preventive maintenance activities

To be competent, the user/individual on the job must be able to:

- PC36.** make sure the machine is kept clean at all times ,before loading, while running & after unloading the fabric
- PC37.** follow the preventive maintenance schedule & ensure the machine is running smoothly
- PC38.** check that all controls are functioning properly
- PC39.** ensure the right quality of water , steam & air is available for proper functioning of machine
- PC40.** cool down the machine after completion of job

Cutting the sample after specified time/ no. of turns

To be competent, the user/individual on the job must be able to:

- PC41.** cut sample as per the time / no. of turn mentioned in the work order to avoid any delay

Qualification Pack

PC42. cut proper size of the sample

PC43. cut the sample from same place every time

PC44. clean hands or cutting tools to avoid soiling of the sample

Developing , soaping, washing & drying the sample

To be competent, the user/individual on the job must be able to:

PC45. develop the sample as per developing techniques for various types of dyestuffs used in the process such as direct, reactive, vat, sulphur etc.

PC46. soap the sample as per SOP for various types of dyestuffs

PC47. dry the sample at clean & dust free place

Comparing the sample with the standard

To be competent, the user/individual on the job must be able to:

PC48. compare the sample with the standard attached to the work order or in the master file

PC49. take the sample to your supervisor if the sample is not matching to the standard for decision

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. standard operating procedure (SOP) and regulations in a processing unit

KU2. safe working practices to be adopted in processing unit

KU3. quality systems and other processes practiced in the processing unit

KU4. reporting to the supervisor or higher authority in case of emergency

KU5. color coding adopted for different products in the processing unit

KU6. the importance of quality & productivity

KU7. various defects in the fabric due to yarn, weaving or processing

KU8. reasons for various defects in the fabric due to processing & their remedy

KU9. process flow in a processing unit

KU10. material flow in a processing unit

KU11. functions of various controls of the machine

KU12. importance of material handling and types of material handling equipment being used

KU13. importance of cleanliness at workplace

KU14. functions and methodology for operating different material handling equipment

KU15. the functions of various alarm signals

KU16. guidelines for operating the machine

KU17. guidelines for taking charge of shift from previous shift operator

KU18. guidelines for handing over the shift to the next shift operator

KU19. safety procedures to be followed while operating the machine

KU20. procedure followed to get the final output in processing unit

KU21. importance of discipline & punctuality

KU22. importance of take over & handover in achieving quality & productivity

KU23. various types of fiber, yarn and fabric

Qualification Pack

- KU24.** process of scouring, bleaching & dyeing & washing
- KU25.** functioning of various control of machine
- KU26.** knowledge about various controls of machine
- KU27.** knowledge about the operations & functions of jigger dyeing machine
- KU28.** knowledge about injection / dosing according to the depth of the shade
- KU29.** use of various checking tools such as thermometer, ph paper etc.
- KU30.** various dyestuffs used & the time or no. of turns required for exhaustion/ fixation of color.
- KU31.** effect of chemicals/ detergents , time & temp of soaping /washing on the dyed fabric

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and short sentences
- GS2.** read and comprehend written instructions
- GS3.** communicate with supervisor appropriately
- GS4.** talk to others to convey information effectively
- GS5.** apply problem-solving approaches in different situations
- GS6.** refer anomalies to the supervisor
- GS7.** apply good attention to detail

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Taking charge of shift from Jigger machine operator</i>	9	10	-	11
PC1. come at least 10 - 15 minutes earlier to the work place	-	-	-	-
PC2. bring the necessary operational tools to the department	-	-	-	-
PC3. meet the previous shift operator and discuss with him/ her regarding the issues faced by him with respect to the quality, production, spare, safety or any other specific instruction etc	-	-	-	-
PC4. understand the fabric being processed & process running on the machine	-	-	-	-
PC5. ensure the technical details are mentioned on the job card display on the machine	-	-	-	-
PC6. check for the availability of the spare trolley/batching trolley for unloading the fabric	-	-	-	-
PC7. check the next batch to be processed is ready near the machine	-	-	-	-
PC8. ensure the required dyes & chemicals are already weighed & prepared	-	-	-	-
PC9. check the cleanliness of the machines & other work areas	-	-	-	-
PC10. question the previous shift operator for any deviation in the above and bring the same to the knowledge of his/ her shift superior as well as that of the previous shift	-	-	-	-
PC11. take over the shift from the outgoing operator in a proper manner	-	-	-	-
<i>Handing Over Shift to Jigger machine operator</i>	9	10	-	11
PC12. ensure proper communication regarding fabric quality & the process running on the machine while providing the details	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. provide all relevant information regarding the stoppages or breakdown in the machine, any damage to the fabric or machine	-	-	-	-
PC14. ensure the empty trolley is near the machine for unloading the fabric	-	-	-	-
PC15. ensure the next lot to be processed, which is already stitched & arranged properly is ready near the machine	-	-	-	-
PC16. ensure the required dyes & chemicals for the next lot or next process are weighed & prepared	-	-	-	-
PC17. get clearance from the incoming counterpart before leaving the work spot	-	-	-	-
PC18. report to his/ her shift superiors as well as that of the incoming shift operator in case his/ her counterpart doesn't report for the incoming shift	-	-	-	-
PC19. ensure ,the shift has to be properly handed over to the incoming shift operator	-	-	-	-
PC20. report to his/ her shift superior about the quality / production / safety issues/ any other issue faced in his/ her shift and leave the department only after getting concurrence for the same from his/ her superiors	-	-	-	-
PC21. collect the wastes from waste collection bags, weigh them and transport to storage area	-	-	-	-
PC22. ensure the machine and its work place is clean	-	-	-	-
<i>Carry out preparatory activities for operations in the machine</i>	50	110	-	5
PC23. understand the task mentioned in the work order	-	-	-	-
PC24. ensure that the machine is empty & clean	-	-	-	-
PC25. load the fabric from the batch trolley in the center of the machine	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. ensure fabric is crease-less and evenly fed onto the machine	-	-	-	-
PC27. prepare the chemical/dye-stuff required for the batch	-	-	-	-
PC28. inject the chemical in the machine tank/dispenser	-	-	-	-
<i>Operate the machine for specified tasks as per Work Order</i>	50	110	-	5
PC29. know the operations of the machine & the plc/ control mechanism programs	-	-	-	-
PC30. read & understand the process being followed to do the task	-	-	-	-
PC31. check the ph, temp & other parameters during the process	-	-	-	-
PC32. maintain even temp across the width of machine	-	-	-	-
PC33. maintain even & minimum tension on the fabric	-	-	-	-
PC34. drain the machine once sample is approved	-	-	-	-
PC35. unload the fabric from the machine after completion of the process	-	-	-	-
<i>Clean the machine on a regular basis and carryout preventive maintenance activities</i>	22	43	-	5
PC36. make sure the machine is kept clean at all times ,before loading, while running & after unloading the fabric	-	-	-	-
PC37. follow the preventive maintenance schedule & ensure the machine is running smoothly	-	-	-	-
PC38. check that all controls are functioning properly	-	-	-	-
PC39. ensure the right quality of water , steam & air is available for proper functioning of machine	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC40. cool down the machine after completion of job	-	-	-	-
<i>Cutting the sample after specified time/ no. of turns</i>	20	43	-	5
PC41. cut sample as per the time / no. of turn mentioned in the work order to avoid any delay	-	-	-	-
PC42. cut proper size of the sample	-	-	-	-
PC43. cut the sample from same place every time	-	-	-	-
PC44. clean hands or cutting tools to avoid soiling of the sample	-	-	-	-
<i>Developing , soaping, washing & drying the sample</i>	20	43	-	5
PC45. develop the sample as per developing techniques for various types of dyestuffs used in the process such as direct, reactive, vat, sulphur etc.	-	-	-	-
PC46. soap the sample as per SOP for various types of dyestuffs	-	-	-	-
PC47. dry the sample at clean & dust free place	-	-	-	-
<i>Comparing the sample with the standard</i>	20	41	-	3
PC48. compare the sample with the standard attached to the work order or in the master file	-	-	-	-
PC49. take the sample to your supervisor if the sample is not matching to the standard for decision	-	-	-	-
NOS Total	200	410	-	50

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TSC/N5220
NOS Name	Carryout shift change, production and quality analysis at jigger machine department
Sector	Textile
Sub-Sector	Processing - Textiles
Occupation	Dyeing & Printing
NSQF Level	2
Credits	7
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TSC/N5221: Carryout shift change, production and quality analysis at soft flow dyeing department

Description

This unit is about carryout shift change, production and quality analysis at soft flow dyeing department

Scope

The scope covers the following :

- taking charge of shift from Soft Flow Dyeing machine operator
- handing over shift to Soft Flow Dyeing machine operator
- carry out preparatory activities for operations in the machine
- operate the machine for the specified tasks as per work order
- clean the machine on a regular basis and carryout preventive maintenance activities
- cutting the sample
- developing ,soaping, washing & drying the sample
- comparing the shade with the standard sample

Elements and Performance Criteria

Taking charge of shift from Soft Flow Dyeing machine operator

To be competent, the user/individual on the job must be able to:

- PC1.** report at least 10 - 15 minutes earlier to the work place and meet the previous shift soft flow dyeing machine operator and discuss with him/ her regarding the issues faced by them with respect to the quality, production, spare, safety or any other specific instruction etc.
- PC2.** ensure to bring the necessary operational tools to the department
- PC3.** understand the fabric being processed and the process running on the machine
- PC4.** ensure the technical details are mentioned on the job card display on the machine
- PC5.** check for the availability of the spare trolley for unloading the fabric
- PC6.** ensure the next batch to be processed is ready near the machine
- PC7.** ensure the required dyes and chemicals are already weighed and prepared
- PC8.** check the cleanliness of the machines and other work areas
- PC9.** obtain information from the previous shift operator for any deviation in the above and bring the same to the knowledge of your shift superior as well that of the previous shift
- PC10.** take charge of the shift from the outgoing Soft Flow Dyeing machine operator in a proper manner

Handing over shift to Soft Flow Dyeing machine operator

To be competent, the user/individual on the job must be able to:

- PC11.** provide all the details regarding fabric quality and the process running on the machine
- PC12.** inform regarding the stoppages or breakdown in the machine, any damage to the fabric or machine
- PC13.** ensure the empty trolley is near the machine for unloading the fabric

Qualification Pack

- PC14.** check that the next lot to be processed is ready near the machine already stitched and arranged properly
- PC15.** get clearance from the incoming counterpart before leaving the work spot
- PC16.** report to your shift superiors as well as that of the incoming shift operator in case your counterpart doesn't report for the incoming shift
- PC17.** report to your shift superior about the quality / production / safety issues/ any other issue faced in your shift and should leave the department only after getting concurrence for the same from your superiors
- PC18.** collect the wastes from waste collection bags, weigh them and transport to storage area
- PC19.** ensure the machine and its work place is clean

Carry out preparatory activities for operations in the machine

To be competent, the user/individual on the job must be able to:

- PC20.** understand the task mentioned in the work order
- PC21.** load the fabric from the batch/trolley in all chambers of the machine equally
- PC22.** fill required quantity of water & close all safety doors properly
- PC23.** prepare the chemical/dye-stuff required for the batch and inject the chemical in the machine tank/dispenser

Operate the machine for specified tasks as per Work Order

To be competent, the user/individual on the job must be able to:

- PC24.** check the pH temperature and other parameters during the process
- PC25.** maintain even temperature across the width of machine
- PC26.** maintain even and minimum tension on the fabric
- PC27.** drain the machine once sample is approved
- PC28.** unload the fabric from the machine after completion of the process

Clean the machine on a regular basis and carryout preventive maintenance activities

To be competent, the user/individual on the job must be able to:

- PC29.** ensure that the machine is kept clean at all times, before loading, while running and after unloading the fabric
- PC30.** follow the preventive maintenance schedule for the machine to run smoothly
- PC31.** check that all controls are functioning properly
- PC32.** ensure the right quality of water, steam and air is available

Cutting the sample

To be competent, the user/individual on the job must be able to:

- PC33.** cut proper size of sample as per the time / no. of turn mentioned in the work order to avoid any delay
- PC34.** ensure to cut the sample from same place every time
- PC35.** clean hands or cutting tools to avoid soiling of the sample

Developing , soaping, washing & drying the sample

To be competent, the user/individual on the job must be able to:

- PC36.** develop the sample as per standard techniques for various types of dyestuffs used in the process such as direct, reactive, vat, sulphur etc

Qualification Pack

PC37. soap the sample using soaping chemical, temp and no. of soaping/ washings as specified for different dyestuff

PC38. dry the sample at a clean and dust free place

Comparing the shade with the standard sample

To be competent, the user/individual on the job must be able to:

PC39. compare the sample with the standard attached to the work order or in the master file

PC40. take the sample to your supervisor if the sample is not matching to the standard for decision

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. standard operating procedure (SOP) and regulations in processing unit

KU2. procedure followed to get the final output in processing unit

KU3. safe working practices to be adopted in processing unit

KU4. reporting method to the supervisor or higher authority about any grievances faced

KU5. the importance of quality & productivity

KU6. various defects in the fabric due to yarn, weaving or processing

KU7. reasons for various defects in the fabric due to processing and, their remedy

KU8. process flow in a processing unit

KU9. material flow in a processing unit

KU10. functions of various controls of the machine

KU11. importance of material handling and types of material handling equipment being used

KU12. importance of cleanliness at workplace

KU13. functions and methodology for operating different material handling equipment

KU14. the functions of various alarm signals

KU15. guidelines for operating the machine

KU16. guidelines for taking charge of shift from previous shift Soft Flow Dyeing machine operator

KU17. guidelines for handing over the shift to the next shift Soft Flow Dyeing machine operator

KU18. safety procedures to be followed while operating the machine

KU19. importance of discipline & punctuality

KU20. process of scouring, bleaching & dyeing

KU21. SOP of dyeing reactive, vat, disperse, direct & other dyestuffs

KU22. effects of various parameters on dyed fabric

KU23. effects of water hardness on fabric dyeing

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. comprehend written instructions

GS2. read any application sent by other colleagues

Qualification Pack

- GS3.** communicate in local language orally
- GS4.** communicate with supervisor appropriately
- GS5.** talk to others to convey information effectively
- GS6.** refer anomalies to the supervisor
- GS7.** apply good attention to detail

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Taking charge of shift from Soft Flow Dyeing machine operator</i>	9	10	-	11
PC1. report at least 10 - 15 minutes earlier to the work place and meet the previous shift soft flow dyeing machine operator and discuss with him/ her regarding the issues faced by them with respect to the quality, production, spare, safety or any other specific instruction etc.	-	-	-	-
PC2. ensure to bring the necessary operational tools to the department	-	-	-	-
PC3. understand the fabric being processed and the process running on the machine	-	-	-	-
PC4. ensure the technical details are mentioned on the job card display on the machine	-	-	-	-
PC5. check for the availability of the spare trolley for unloading the fabric	-	-	-	-
PC6. ensure the next batch to be processed is ready near the machine	-	-	-	-
PC7. ensure the required dyes and chemicals are already weighed and prepared	-	-	-	-
PC8. check the cleanliness of the machines and other work areas	-	-	-	-
PC9. obtain information from the previous shift operator for any deviation in the above and bring the same to the knowledge of your shift superior as well that of the previous shift	-	-	-	-
PC10. take charge of the shift from the outgoing Soft Flow Dyeing machine operator in a proper manner	-	-	-	-
<i>Handing over shift to Soft Flow Dyeing machine operator</i>	9	10	-	11
PC11. provide all the details regarding fabric quality and the process running on the machine	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. inform regarding the stoppages or breakdown in the machine, any damage to the fabric or machine	-	-	-	-
PC13. ensure the empty trolley is near the machine for unloading the fabric	-	-	-	-
PC14. check that the next lot to be processed is ready near the machine already stitched and arranged properly	-	-	-	-
PC15. get clearance from the incoming counterpart before leaving the work spot	-	-	-	-
PC16. report to your shift superiors as well as that of the incoming shift operator in case your counterpart doesn't report for the incoming shift	-	-	-	-
PC17. report to your shift superior about the quality / production / safety issues/ any other issue faced in your shift and should leave the department only after getting concurrence for the same from your superiors	-	-	-	-
PC18. collect the wastes from waste collection bags, weigh them and transport to storage area	-	-	-	-
PC19. ensure the machine and its work place is clean	-	-	-	-
<i>Carry out preparatory activities for operations in the machine</i>	51	110	-	5
PC20. understand the task mentioned in the work order	-	-	-	-
PC21. load the fabric from the batch/trolley in all chambers of the machine equally	-	-	-	-
PC22. fill required quantity of water & close all safety doors properly	-	-	-	-
PC23. prepare the chemical/dye-stuff required for the batch and inject the chemical in the machine tank/dispenser	-	-	-	-
<i>Operate the machine for specified tasks as per Work Order</i>	51	110	-	5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. check the pH temperature and other parameters during the process	-	-	-	-
PC25. maintain even temperature across the width of machine	-	-	-	-
PC26. maintain even and minimum tension on the fabric	-	-	-	-
PC27. drain the machine once sample is approved	-	-	-	-
PC28. unload the fabric from the machine after completion of the process	-	-	-	-
<i>Clean the machine on a regular basis and carryout preventive maintenance activities</i>	20	44	-	5
PC29. ensure that the machine is kept clean at all times, before loading, while running and after unloading the fabric	-	-	-	-
PC30. follow the preventive maintenance schedule for the machine to run smoothly	-	-	-	-
PC31. check that all controls are functioning properly	-	-	-	-
PC32. ensure the right quality of water, steam and air is available	-	-	-	-
<i>Cutting the sample</i>	20	44	-	5
PC33. cut proper size of sample as per the time / no. of turn mentioned in the work order to avoid any delay	-	-	-	-
PC34. ensure to cut the sample from same place every time	-	-	-	-
PC35. clean hands or cutting tools to avoid soiling of the sample	-	-	-	-
<i>Developing , soaping, washing & drying the sample</i>	20	42	-	5
PC36. develop the sample as per standard techniques for various types of dyestuffs used in the process such as direct, reactive, vat, sulphur etc	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC37. soap the sample using soaping chemical, temp and no. of soaping/ washings as specified for different dyestuff	-	-	-	-
PC38. dry the sample at a clean and dust free place	-	-	-	-
<i>Comparing the shade with the standard sample</i>	20	40	-	3
PC39. compare the sample with the standard attached to the work order or in the master file	-	-	-	-
PC40. take the sample to your supervisor if the sample is not matching to the standard for decision	-	-	-	-
NOS Total	200	410	-	50

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TSC/N5221
NOS Name	Carryout shift change, production and quality analysis at soft flow dyeing department
Sector	Textile
Sub-Sector	Processing - Textiles
Occupation	Dyeing & Printing
NSQF Level	2
Credits	7
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TSC/N5222: Carryout shift change, production and quality analysis at package dyeing department

Description

This unit is about carryout shift change, production and quality analysis at package dyeing department

Scope

The scope covers the following :

- taking charge of shift from Package Dyeing machine operator
- handing over shift to Package Dyeing machine operator
- carry out preparatory activities for operations in the machine
- operate the machine for specified task as per work order
- clean the machine on a regular basis & carryout preventive maintenance
- taking out the sample after specified hold time
- developing, soaping, washing & drying the sample
- comparing the sample with the standard

Elements and Performance Criteria

Taking charge of shift from Package Dyeing machine operator

To be competent, the user/individual on the job must be able to:

- PC1.** come at least 10 - 15 minutes earlier to the work place
- PC2.** bring the necessary operational tools to the department
- PC3.** meet the previous shift operator and discuss with him/ her regarding the issues faced by them with respect to the quality, production, spare, safety or any other specific instruction etc.
- PC4.** understand the fibre / yarn being processed & process running on the machine
- PC5.** ensure the technical details are mentioned on the job card display on the machine
- PC6.** should check for the availability of the spare trolley for unloading the yarn packages
- PC7.** should check the next batch to be processed is ready near the machine
- PC8.** must ensure the required dyes & chemicals are already weighed & prepared
- PC9.** should check the cleanliness of the machines & other work areas
- PC10.** question the previous shift operator for any deviation in the above and should bring the same to the knowledge of his/ her shift superior as well as that of the previous shift.
- PC11.** take over the shift from the outgoing operator in a proper manner

Handing over shift to Package Dyeing machine operator

To be competent, the user/individual on the job must be able to:

- PC12.** ensure in providing the details regarding yarn quality & the process running on the machine
- PC13.** provide all relevant information regarding the stoppages or breakdown in the machine, any damage to the yarn or machine.
- PC14.** ensure the empty trolley is near the machine for unloading the yarn packages

Qualification Pack

- PC15.** ensure the next lot to be processed is ready near the machine
- PC16.** ensure the required dyes & chemicals for the next lot or next process are weighed & prepared
- PC17.** get clearance from the incoming counterpart before leaving the work spot
- PC18.** report to his/ her shift superiors as well as that of the incoming shift operator in case his/ her counterpart doesn't report for the incoming shift
- PC19.** ensure the shift has to be properly handed over to the incoming shift operator
- PC20.** report to his/ her shift superior about the quality / production / safety issues/ any other issue faced in his/ her shift and should leave the department only after getting concurrence for the same from his/ her superiors
- PC21.** collect the wastes from waste collection bags, weigh them and transport to storage area
- PC22.** ensure the machine and its work place is clean

Carry out preparatory activities for operations in the machine

To be competent, the user/individual on the job must be able to:

- PC23.** understand the task mentioned in the work order
- PC24.** ensure that the machine is empty & clean
- PC25.** receiving the material to be processed
- PC26.** loading the material to be processed in the carrier
- PC27.** closing the machine lid & ensuring the safety locks are in place
- PC28.** fill & start the machine as per specified MLR
- PC29.** prepare & inject the dyes / chemicals in the machine

Operate the machine for specified tasks as per Work Order

To be competent, the user/individual on the job must be able to:

- PC30.** know the operations of the machine & the plc/control mechanism programs
- PC31.** read & understand the process being followed to do the task
- PC32.** check the ph, temp & other parameters during the process
- PC33.** attend to the machine signals for manual intervention
- PC34.** ensure the rate of heating/ cooling, final temp & hold time is as programmed

Clean the machine on a regular basis and carryout preventive maintenance activities

To be competent, the user/individual on the job must be able to:

- PC35.** make sure the machine is kept clean at all times ,before loading, while running & after unloading the fabric
- PC36.** follow the preventive maintenance schedule & ensure the machine is running smoothly
- PC37.** ensure the right quality of water, steam & air is available

Taking out the sample after specified hold time

To be competent, the user/individual on the job must be able to:

- PC38.** take out the sample after hold time to avoid any delay
- PC39.** close the lid & restart the machine after drawing the sample

Developing , soaping, washing & drying the sample

To be competent, the user/individual on the job must be able to:

Qualification Pack

PC40. develop the sample as per SOP techniques for various types of dyestuffs used in the process such as direct, reactive, vat, sulphur etc.

PC41. soap the sample as per SOP for soaping for various dyestuffs

PC42. dry the sample at clean & dust free place

Comparing the shade with the standard sample

To be competent, the user/individual on the job must be able to:

PC43. compare the sample with the standard attached to the work order or in the master file

PC44. take the sample to your supervisor if the sample is not matching to the standard for decision

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. standard operating procedure (SOP) and regulations in processing unit

KU2. procedure followed to get the final output in processing unit

KU3. safe working practices to be adopted in processing unit

KU4. reporting method to the supervisor or higher authority about any grievances faced

KU5. the importance of quality & productivity

KU6. various defects in the yarn due to spinning or processing

KU7. reasons for various defects in the yarn due to processing & their remedy

KU8. process flow in a processing unit

KU9. material flow in a processing unit

KU10. functions of various controls of the machine

KU11. importance of material handling and types of material handling equipment being use

KU12. importance of cleanliness at workplace

KU13. functions and methodology for operating different material handling equipment

KU14. the functions of various alarm signals

KU15. guidelines for operating the machine

KU16. guidelines for taking charge of shift from previous shift Package Dyeing machine operator

KU17. guidelines for handing over the shift to the next shift Package Dyeing machine operator

KU18. safety procedures to be followed while operating the machine

KU19. importance of discipline & punctuality

KU20. process of scouring, bleaching & dyeing

KU21. knowledge about package weight & density

KU22. operation of pneumatic & hydraulic press

KU23. knowledge of injection/ dosing techniques

KU24. operation of fill, drain, heating, cooling valves

KU25. effects of various parameters such as time , temp, pH etc on dye

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** comprehend written instructions
- GS2.** read any application sent by other colleagues
- GS3.** communicate in simple language orally
- GS4.** communicate with supervisor appropriately
- GS5.** talk to others to convey information effectively
- GS6.** identify the real reason of problem faced
- GS7.** apply problem-solving approaches in different situations
- GS8.** refer anomalies to the supervisor
- GS9.** seek clarification on problems from others
- GS10.** apply good attention to detail

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Taking charge of shift from Package Dyeing machine operator</i>	9	10	-	11
PC1. come at least 10 - 15 minutes earlier to the work place	-	-	-	-
PC2. bring the necessary operational tools to the department	-	-	-	-
PC3. meet the previous shift operator and discuss with him/ her regarding the issues faced by them with respect to the quality, production, spare, safety or any other specific instruction etc.	-	-	-	-
PC4. understand the fibre / yarn being processed & process running on the machine	-	-	-	-
PC5. ensure the technical details are mentioned on the job card display on the machine	-	-	-	-
PC6. should check for the availability of the spare trolley for unloading the yarn packages	-	-	-	-
PC7. should check the next batch to be processed is ready near the machine	-	-	-	-
PC8. must ensure the required dyes & chemicals are already weighed & prepared	-	-	-	-
PC9. should check the cleanliness of the machines & other work areas	-	-	-	-
PC10. question the previous shift operator for any deviation in the above and should bring the same to the knowledge of his/ her shift superior as well as that of the previous shift.	-	-	-	-
PC11. take over the shift from the outgoing operator in a proper manner	-	-	-	-
<i>Handing over shift to Package Dyeing machine operator</i>	9	10	-	11
PC12. ensure in providing the details regarding yarn quality & the process running on the machine	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. provide all relevant information regarding the stoppages or breakdown in the machine, any damage to the yarn or machine.	-	-	-	-
PC14. ensure the empty trolley is near the machine for unloading the yarn packages	-	-	-	-
PC15. ensure the next lot to be processed is ready near the machine	-	-	-	-
PC16. ensure the required dyes & chemicals for the next lot or next process are weighed & prepared	-	-	-	-
PC17. get clearance from the incoming counterpart before leaving the work spot	-	-	-	-
PC18. report to his/ her shift superiors as well as that of the incoming shift operator in case his/ her counterpart doesn't report for the incoming shift	-	-	-	-
PC19. ensure the shift has to be properly handed over to the incoming shift operator	-	-	-	-
PC20. report to his/ her shift superior about the quality / production / safety issues/ any other issue faced in his/ her shift and should leave the department only after getting concurrence for the same from his/ her superiors	-	-	-	-
PC21. collect the wastes from waste collection bags, weigh them and transport to storage area	-	-	-	-
PC22. ensure the machine and its work place is clean	-	-	-	-
<i>Carry out preparatory activities for operations in the machine</i>	50	110	-	5
PC23. understand the task mentioned in the work order	-	-	-	-
PC24. ensure that the machine is empty & clean	-	-	-	-
PC25. receiving the material to be processed	-	-	-	-
PC26. loading the material to be processed in the carrier	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. closing the machine lid & ensuring the safety locks are in place	-	-	-	-
PC28. fill & start the machine as per specified MLR	-	-	-	-
PC29. prepare & inject the dyes / chemicals in the machine	-	-	-	-
<i>Operate the machine for specified tasks as per Work Order</i>	50	110	-	5
PC30. know the operations of the machine & the plc/control mechanism programs	-	-	-	-
PC31. read & understand the process being followed to do the task	-	-	-	-
PC32. check the ph, temp & other parameters during the process	-	-	-	-
PC33. attend to the machine signals for manual intervention	-	-	-	-
PC34. ensure the rate of heating/ cooling, final temp & hold time is as programmed	-	-	-	-
<i>Clean the machine on a regular basis and carryout preventive maintenance activities</i>	22	43	-	5
PC35. make sure the machine is kept clean at all times ,before loading, while running & after unloading the fabric	-	-	-	-
PC36. follow the preventive maintenance schedule & ensure the machine is running smoothly	-	-	-	-
PC37. ensure the right quality of water, steam & air is available	-	-	-	-
<i>Taking out the sample after specified hold time</i>	20	43	-	5
PC38. take out the sample after hold time to avoid any delay	-	-	-	-
PC39. close the lid & restart the machine after drawing the sample	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Developing , soaping, washing & drying the sample</i>	20	42	-	5
PC40. develop the sample as per SOP techniques for various types of dyestuffs used in the process such as direct, reactive, vat, sulphur etc.	-	-	-	-
PC41. soap the sample as per SOP for soaping for various dyestuffs	-	-	-	-
PC42. dry the sample at clean & dust free place	-	-	-	-
<i>Comparing the shade with the standard sample</i>	20	42	-	3
PC43. compare the sample with the standard attached to the work order or in the master file	-	-	-	-
PC44. take the sample to your supervisor if the sample is not matching to the standard for decision	-	-	-	-
NOS Total	200	410	-	50

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TSC/N5222
NOS Name	Carryout shift change, production and quality analysis at package dyeing department
Sector	Textile
Sub-Sector	Processing - Textiles
Occupation	Dyeing & Printing
NSQF Level	2
Credits	7
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

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(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N9015.Follow machine, safety, and organizational guidelines in textile sector	65	80	-	50	195	20
TSC/N9016.Follow teamwork, adaptability, and communication guidelines in textile sector	15	20	-	10	45	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	100	130	-	60	290	30

Elective: 1 Jigger

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N5220.Carryout shift change,production and quality analysis at jigger machine department	200	410	-	50	660	70
Total	200	410	-	50	660	70

Elective: 2 Soft Flow Dyeing

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N5221.Carryout shift change, production and quality analysis at soft flow dyeing department	200	410	-	50	660	70
Total	200	410	-	50	660	70

Elective: 3 Package Dyeing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N5222.Carryout shift change,production and quality analysis at package dyeing department	200	410	-	50	660	70
Total	200	410	-	50	660	70

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.